

Course: BCSNT 6063 – Object Oriented Programming (2nd Semester)

Assignment Type: Practical / Tutorial

Submission Date:: 6th June 2026 (Saturday)

1) Smart Attendance Checker

```
import java.util.Scanner;

public class AttendanceChecker{
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);

        System.out.print("Enter num of class conducted:");
        int classConducted=sc.nextInt();
        System.out.print("Enter num of class attended:");
        int classAttended= sc.nextInt();

        if(classConducted <= 0){
            system.out.println("Invalid Input");
        }else{
            double attendance = ((double)classAttended / classConducted)
                                * 100;
            System.out.printf("Attendance Percentage: %.2f",attendance);

            if(attendance >= 75){
                system.out.println("Eligible");
            }else{
                System.out.println("Not Eligible");
            }
            sc.close();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\j
Enter number of classes conducted: 35
Enter number of classes attended: 28
Attendance Percentage: 80.00
Eligible for Exam

Process finished with exit code 0
```

2) AI Age Predictor

```
import java.util.Scanner;

public class AgePredictor{
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);
        System.out.print("Enter your age:");
        int age=sc.nextInt();
        int age10 = age+10;
        int age25 = age+25;
        int age50 = age+50;
        int age100 = 100-age;
        System.out.println("\n--- Age Predictor ---");
        System.out.println("Age after 10 yrs=" + age10);
        System.out.println("Age after 25 yrs=" + age25);
        System.out.println("Age after 50 yrs=" + age50);
        System.out.println("You will be 100 in " + yearage100);
        sc.close();
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\
Enter your current age: 19

--- Age Predictions ---
Age after 10 years: 29
Age after 25 years: 44
Age after 50 years: 69
You will turn 100 after 81years.

Process finished with exit code 0
```

3) Fitness Challenge Tracker

```
import java.util.Scanner;

public class FitnessTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] steps = new int[7];
        int total = 0;
        int highest = 0;

        System.out.print("Enter steps walked in 7 days:");
        for (int i = 0; i < 7; i++) {
            System.out.print("Day " + (i + 1) + ": ");
            steps[i] = sc.nextInt();

            total += steps[i];
            if (steps[i] > highest) {
                highest = steps[i];
            }
        }
        double avg = (double) total / 7;

        System.out.println("Total Steps: " + total);
        System.out.printf("Avg steps: %.2f\n", avg);
        System.out.printf("Highest step: " + highest);

        sc.close();
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\ja
Enter steps walked for 7 days:
Day 1: 1000
Day 2: 2000
Day 3: 3000
Day 4: 4000
Day 5: 500
Day 6: 6000
Day 7: 800

<--- Fitness Report --->
Total steps: 17300
Average steps: 2471.43
Highest steps in a day: 6000

Process finished with exit code 0
```

4) Fibonacci Series

```
import java.util.Scanner;

public class Fibonacci {
    public static int fib(int n) {
        if (n == 0) return 0;
        if (n == 1) return 1;
        return fib(n-1) + fib(n-2);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of terms = ");
        int n = sc.nextInt();

        System.out.println("\n Fibonacci Series \n");

        for (int i = 0; i < n; i++) {
            System.out.print(fib(i) + " ");
        }

        sc.close();
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:
Enter number of terms (n): 19

Fibonacci Series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584
Process finished with exit code 0
```

5) Prime Number Checker

```
import java.util.Scanner;

public class Prime{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        System.out.print("Enter a number:");
        int n=sc.nextInt();
        boolean prime=true;

        if(n<=1) prime=false;

        for (int i=2; i<= Math.sqrt(n); i++){
            if(n%i==0){
                prime=false;
                break;
            }
        }

        System.out.print(n+" prime? "Prime Number": "Not
            Prime Number");
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"
Enter a number: 77
Not Prime Number

Process finished with exit code 0
```

6) Logical Operators Check

```
import java.util.Scanner;

public class Logic {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter three integer: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if ((a+b) == c) {
            System.out.println("Third num is sum of two");
        } else {
            System.out.println("Not true");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\ja
Enter three integer numbers : 10 20 30
Third number is the sum of first two.

Process finished with exit code 0
```

7) Multiplication Table

```
import java.util.Scanner;

public class Table{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        System.out.print("Enter a number:");
        int n=sc.nextInt();
        System.out.println("Table");
        for(int i=1; i<=10; i++){
            System.out.printf("%d x %d= %d\n", n,i,(n*i));
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin
Enter a number: 28
<----- Multiplication Table ----->
28 x 1 = 28
28 x 2 = 56
28 x 3 = 84
28 x 4 = 112
28 x 5 = 140
28 x 6 = 168
28 x 7 = 196
28 x 8 = 224
28 x 9 = 252
28 x 10 = 280
<----->

Process finished with exit code 0
```

8) Right-Angled Triangle Pattern

```
import java.util.Scanner

public class Triangle{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        System.out.print("Enter num of rows:");
        int rows= sc.nextInt();
        for(int i=1; i<=rows; i++){
            for(int j=1; j<=i; j++){
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\
Enter number of rows: 5
*
**
***
****
*****

Process finished with exit code 0
|
```

9)Pyramid Pattern

```
import java.util.Scanner;

public class Pyramid {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter no. of rows: ");
        int rows = sc.nextInt();

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= rows - i; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2 * i - 1); j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

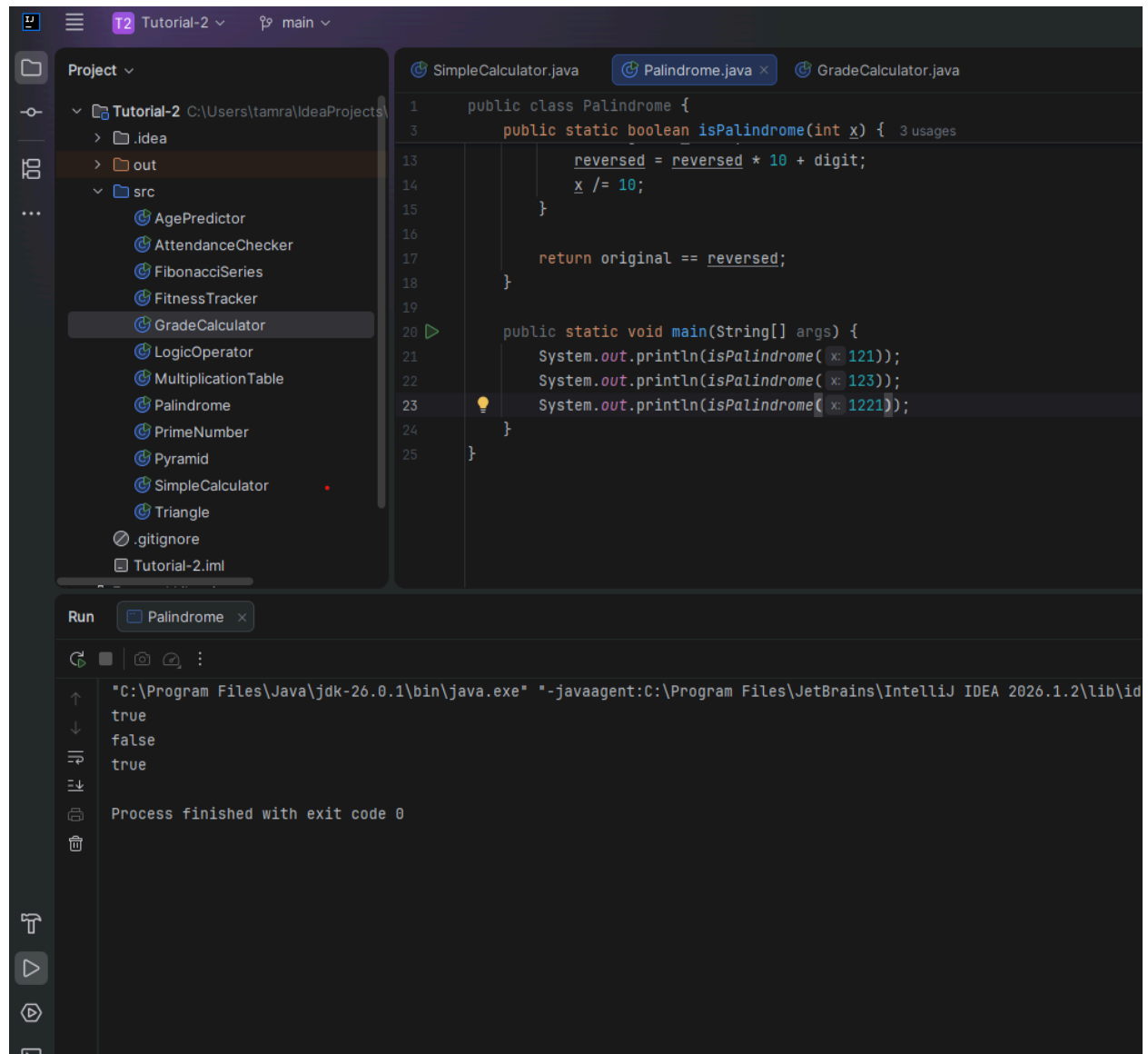
```
"C:\Program Files\Java\jdk-26.0.1
Enter number of rows: 5
  *
 ***
*****
*****
*****

Process finished with exit code 0
```

10) Palindrome Number

```
public class Palindrome {  
    public static boolean isPalindrome(int x) {  
        if (x < 0) {  
            return false;  
        }  
        int original original = x;  
        int reversed = 0;  
  
        while (x != 0) {  
            int digit = x % 10;  
            reversed = reversed * 10 + digit;  
            x /= 10;  
        }  
        return original == true reversed;  
    }  
  
    public static void main(String[] args) {  
        System.out.println(isPalindrome(1221));  
    }  
}
```

Output:



The screenshot shows the IntelliJ IDEA interface. On the left, the 'Project' view displays the file structure of 'Tutorial-2', with 'src/Palindrome.java' selected. The main editor window shows the code for the 'Palindrome' class. The code defines a static method 'isPalindrome' that takes an integer 'x' and returns a boolean. It uses a loop to reverse the digits of 'x' and then compares the reversed value to the original. The 'main' method calls 'isPalindrome' with the values 121, 123, and 1221. Below the code, the 'Run' tab shows the execution output, which displays 'true', 'false', and 'true' for the three inputs respectively. The process finished with exit code 0.

```
1 public class Palindrome {
2     public static boolean isPalindrome(int x) {
3         3 usages
4         reversed = reversed * 10 + digit;
5         x /= 10;
6     }
7     return original == reversed;
8 }
9
10 public static void main(String[] args) {
11     System.out.println(isPalindrome(x: 121));
12     System.out.println(isPalindrome(x: 123));
13     System.out.println(isPalindrome(x: 1221));
14 }
15 }
```

Run Palindrome x

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.2\lib\id
true
false
true
Process finished with exit code 0

11) Simple Calculator

```
import java.util.Scanner;

public class Calc {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter 1st number :");
        double a = sc.nextDouble();

        System.out.print("Enter 2nd number :");
        double b = sc.nextDouble();
    }
}
```

```
System.out.print("Enter operator(+, -, *, /):");  
char operator = sc.nextChar();
```

```
double result = 0;
```

```
switch (operator) {
```

```
    case '+':
```

```
        result = a + b;
```

```
        System.out.println("Result = " + result);
```

```
        break;
```

```
    case '-':
```

```
        result = a - b;
```

```
        System.out.println("Result = " + result);
```

```
        break;
```

```
    case '*':
```

```
        result = a * b;
```

```
        System.out.println("Result = " + result);
```

```
        break;
```

```
    case '/':
```

```
        if (b != 0) {
```

```
            result = a / b;
```

```
            System.out.println("Result = " + result);
```

```
        } else {
```

```
            System.out.println("Error");
```

```
        }
```

```
        break;
```

```
    default: System.out.println("Invalid");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bi
Enter first number: 10
Enter second number: 20
Enter operator (+, -, *, /): /
Result = 0.5

Process finished with exit code 0
```

12)Grade Calculator

```
import java.util.Scanner;

public class Grade{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        System.out.print("Enter marks:");
        int marks = sc.nextInt();

        string grade=
            (marks >= 90)? "A+":
                (mark >= 85)? "A":
                    (mark >= 80)? "A-":
                        (mark >= 75)? "B+":
                            (mark >= 70)? "B":
                                (mark >= 65)? "B-":
                                    (mark >= 60)? "C+":
                                        (mark >= 55)? "C":
                                            (mark >= 50)? "C-":
                                                "F";

        System.out.println("Grade:" + grade);

        sc.close();
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\java
Enter marks: 80
Grade: A-

Process finished with exit code 0
```